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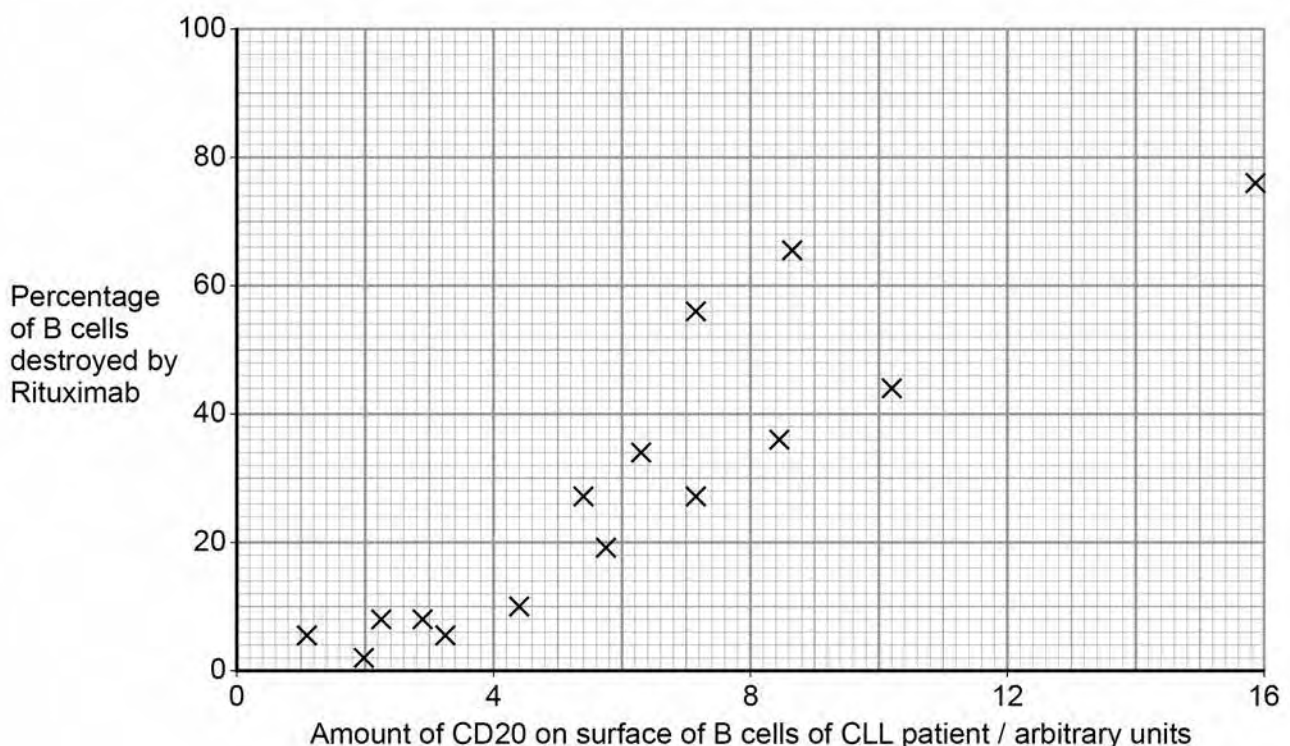
Chronic lymphocytic leukaemia (CLL) is a cancer that affects some B cells of a person's immune system.

Rituximab is a drug used to treat CLL. It binds to a protein called CD20 on the surface of B cells. If enough Rituximab binds to a B cell, it can kill the cell. Rituximab kills **both** healthy **and** cancerous B cells. The body then produces new B cells.

The amount of CD20 on the surface of B cells varies from one person to another. Doctors investigated the relationship between the amount of CD20 on the B cells of a patient and the percentage of B cells destroyed by Rituximab.

Figure 2 shows the doctors' results. Each cross is the result for one patient.

Figure 2



0 2 . 1

What statistical test could the scientists have used to determine whether there was a significant relationship between the amount of CD20 on the surface of B cells and the percentage of B cells destroyed by Rituximab?
Give a reason for your answer.

[1 mark]

Name of test _____

Reason _____

From these data, what can you conclude about the effectiveness of Rituximab in treating patients with CLL?

[3 marks]

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Approximately 11 percent of people with CLL also have mutations of a gene called *NOTCH1*. This leads to production of a non-functional transcription factor associated with CD20 production.

The doctors determined the median percentage of B cells destroyed by Rituximab in people with CLL who had the *NOTCH1* mutation and those who did not.

The doctors' results are shown in **Table 1**.

Table 1

	Median percentage of B cell destroyed by Rituximab
In people with CLL who had the <i>NOTCH1</i> mutation	4
In people with CLL who did not have <i>NOTCH1</i> mutation	22

0 2 . 3 Human blood contains (approximately) 1.0×10^9 B cells per dm^3 .

Use the median values in **Table 1** to calculate the difference between the number of B cells per dm^3 in the blood of people treated with Rituximab with the *NOTCH1* mutation and people without the *NOTCH1* mutation.

Express your answer in standard form. Show your working.

[2 marks]

_____ cells per dm^3

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